

Karissa Palmer

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SUMMARY

Final year PhD student in Statistics with research background in Bayesian methods for spatio-temporal data and experience in education and consulting in applied statistical methods.

EDUCATION

Colorado State University	Fort Collins, CO
PhD Candidate, Statistics	2021 - Present
MS, Statistics	
South Dakota State University	Brookings, SD
MS, Mathematics with Statistics Specialization	2019 - 2021
BA, Spanish	
University of South Dakota	Vermillion, SD
BS, Mathematics, <i>summa cum laude</i>	2015 - 2019

PROFESSIONAL EXPERIENCE

Colorado State University	Fort Collins, CO
Graduate Teaching Assistant	2021 - Present
Graduate Research Assistant (FA 2023 - SP 2024, SU 2025, SU 2026)	
CableLabs	Louisville, CO
Applied Strategist Intern	SU 2021
Analyzed the usage of mobile data to create a predictive model. In addition, researched the relationships between the major cell phone carriers in the United States.	
South Dakota State University	Brookings, SD
CAPITAL Card Services Research Fellow (FA 2020 - SP 2021)	2019 - 2021
Graduate Teaching Assistant	
CAPITAL Card Services	Sioux Falls, SD
Statistician Intern	SU 2019
Created an action-effect model to predict charge-off probability.	
Sanford Health, Enterprise and Data Analytics	Sioux Falls, SD
Advanced Analytics Intern	SU 2018
Used parametric clustering techniques to group patients using demographic data.	
University of South Dakota	Vermillion, SD
Tutor	2016 - 2019

TEACHING EXPERIENCE

Colorado State University	Fort Collins, CO
Instructor	
STAT 100: Statistical Literacy	SP 2026
STAT 201: General Statistics	SP 2026
STAT 301/307: Introduction to Statistical Methods	SP 2022, FA 2022, SP 2023, SU 2024, FA 2025, FA 2026
STAT 315: Introduction to Theory and Practice of Statistics	FA 2024, SP 2025
Undergraduate Research Mentor	
Helped advise summer research project for student on data visualization.	SU 2024
Course Developer	
STAT 307: Introduction to Biostatistics	SU 2023
Grader	
STAR 513: Regression Models for Researchers	SP 2022
STAT 331: Intermediate Applied Statistical Methods	FA 2022
STAT 600: Statistical Computing	FA 2025
Recitation Instructor	
STAT 201: General Statistics	FA 2021
Short Course Instructor/Facilitator	
Coding & Cookies: Introduction to Python	FA 2026
Coding & Cookies: Working with Data in Python	FA 2025
Coding & Cookies: Data Visualization with Python	FA 2024 - SP 2025

South Dakota State University
Graduate Teaching Assistant
MATH 114: College Algebra
MATH 121: Survey of Calculus

Brookings, SD
FA 2019
FA 2019, SP 2020

RESEARCH EXPERIENCE

Bayesian Methods for Spatial Trajectory Data <i>PhD Dissertation</i> Dissertation Advisor: Yawen Guan	2023 - Present
Variable Correlation in Logistic Regression <i>MS Research Paper</i> Advisor: Thomas Brandenburger	2020 - 2021
Honey Bee and Varroa destructor Population Dynamics <i>Undergraduate Honors Thesis</i> Advisor: Dan Van Peurse	2018 - 2019

PUBLICATIONS

In Preparation Palmer, K., Guan, Y., Hammerquist, C., and Sulsky, D. (2026+). A spatial product partition model for clustering sea ice trajectories. To be submitted. Palmer, K., Guan, Y. (2026+). TrajPPM: An R package for trajectory-based product partition models. To be submitted.	
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PRESENTATIONS

ENVR Workshop "A Spatio-Temporal Model for Arctic Sea Ice Trajectories" Poster Presentation	Boulder, CO 10/2024
University of South Dakota IdeaFest "Honey Bee and Varroa destructor Population Dynamics" Student Paper Oral Presentation	Vermillion, SD 04/2019
Mathematical Association of America MathFest "Complex Numbers, Ptolemy's Theorem, and the Law of Cosines" Student Paper Oral Presentation	Chicago, IL 07/2017

AWARDS AND HONORS

Colorado State University Statistics Department Distinguished Service Award Elmer E. Remmenga Scholarship in Applied Statistics Programs of Research and Scholarly Excellence (PRSE) Fellowship	Fort Collins, CO 2026 - 2027 2023 - 2024 2021 - 2022
University of South Dakota Honors Program Thesis with Distinction Dr. Marjorie H. Beaty Math/Computer Science Scholarship NASA Space Grant Educational Stipend, South Dakota Space Grant Consortium	Vermillion, SD 2019 2018 - 2019 2017, 2018, 2020

TECHNICAL SKILLS

Programming R (advanced), C++ via Rcpp (intermediate), Python (basic)	
Relevant Courses Probability Theory, Numerical Analysis, Mathematical Statistics, Advanced Statistical Data Analysis, Bayesian Statistics, Spatial Statistics, Computational Statistics	
Spoken Languages Spanish (limited working proficiency)	